



Data Storm 6.0

Project Report

Team - Calm Coders

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Chapter 1 - Exploratory Data Analysis

0 Introduction & Dataset Description

0.1 Purpose of EDA

The purpose of Exploratory Data Analysis (EDA) in this context is to gain a thorough understanding of the agent activity dataset before proceeding to model development. EDA helps identify patterns, trends, and anomalies within the data, assess data quality, and reveal relationships between features and the target variable. It also guides the selection of relevant features, highlights potential data preprocessing needs (such as handling missing values or outliers), and ensures that the dataset is suitable for predictive modeling.

0.2 Goals

The main goals of performing EDA on this dataset are to understand how agent activities correlate with their ability to sell policies, identify the most influential features for predicting NILL status, and detect any structural or quality issues in the data. By analyzing trends across different time windows and agent profiles, we aim to generate insights that inform feature engineering and improve model performance. Ultimately, this analysis supports the development of a reliable classification model to predict which agents are at risk of becoming One Month NILL in the future.

	count	mean	std	min	25%	50%	75%	max	range
agent_age	15308.0	40.59	11.69	20.0	31.00	41.0	51.00	60.0	40.0
unique_proposals_last_7_days	15308.0	1.49	1.12	0.0	0.00	1.0	2.00	3.0	3.0
unique_proposals_last_15_days	15308.0	2.99	2.00	0.0	1.00	3.0	5.00	6.0	6.0
unique_proposals_last_21_days	15308.0	10.05	6.04	0.0	5.00	10.0	15.00	20.0	20.0
unique_proposal	15308.0	17.52	6.62	1.0	12.00	17.0	23.00	34.0	33.0
unique_quotations_last_7_days	15308.0	1.99	1.42	0.0	1.00	2.0	3.00	4.0	4.0
unique_quotations_last_15_days	15308.0	3.00	2.00	0.0	1.00	3.0	5.00	6.0	6.0
unique_quotations_last_21_days	15308.0	4.47	2.88	0.0	2.00	4.0	7.00	9.0	9.0
unique_quotations	15308.0	13.92	4.66	1.0	11.00	14.0	17.00	32.0	31.0
unique_customers_last_7_days	15308.0	3.00	1.64	0.0	2.00	3.0	4.00	6.0	6.0
unique_customers_last_15_days	15308.0	5.00	2.44	0.0	3.00	5.0	7.00	10.0	10.0
unique_customers_last_21_days	15308.0	7.50	3.47	0.0	5.00	8.0	10.00	15.0	15.0
unique_customers	15308.0	15.49	4.54	1.0	12.00	15.0	19.00	31.0	30.0
new_policy_count	15308.0	20.27	9.54	0.0	15.00	21.0	27.00	42.0	42.0
ANBP_value	15308.0	1025337.79	805498.86	0.0	347323.75	903144.5	1560699.50	3933840.0	3933840.0
net_income	15308.0	228041.39	173667.10	1160.0	92825.00	189974.0	321744.75	1140237.0	1139077.0
number_of_policy_holders	15308.0	30.97	24.62	0.0	12.00	28.0	48.00	116.0	116.0
number_of_cash_payment_policies	15308.0	101.18	75.32	0.0	38.00	90.0	152.00	378.0	378.0

Figure 1: Overview

0.3 Feature Definitions

Below is a brief description of each feature in the dataset:

- **Row ID**: Unique identifier for each record (categorical, nominal).
- **agent_code**: Unique code assigned to each agent (categorical, nominal).
- **agent_age**: Age of the agent in years (numerical, integer).
- **agent_join_month**: Month in which the agent joined the company, represented numerically (categorical, ordinal).
- **first_policy_sold_month**: Month when the agent sold their first policy, represented numerically (categorical, ordinal).
- **year_month**: The year and month corresponding to the record, typically in YYYY-MM format (categorical, nominal or datetime, depending on processing).
- **unique_proposals_last_7_days**: Number of unique proposals created in the past 0–7 days (numerical, integer).
- **unique_proposals_last_15_days**: Unique proposals created in days 7–15 prior (numerical, integer).
- **unique_proposals_last_21_days**: Unique proposals created in days 15–21 prior (numerical, integer).
- **unique_proposal**: Total number of proposals created in the entire month (numerical, integer).
- **unique_quotations_last_7_days**: Quotations generated in the last 0–7 days (numerical, integer).
- **unique_quotations_last_15_days**: Quotations generated in days 7–15 prior (numerical, integer).
- **unique_quotations_last_21_days**: Quotations generated in days 15–21 prior (numerical, integer).
- **unique_quotations**: Total number of quotations in the month (numerical, integer).
- **unique_customers_last_7_days**: Customers contacted in the last 0–7 days (numerical, integer).
- **unique_customers_last_15_days**: Customers contacted in days 7–15 prior (numerical, integer).
- **unique_customers_last_21_days**: Customers contacted in days 15–21 prior (numerical, integer).
- **unique_customers**: Total unique customers contacted in the month (numerical, integer).
- **new_policy_count**: Number of new insurance policies sold during the recorded month (numerical, integer).

- **ANBP_value**: Annualized New Business Premium value in monetary units, representing the annual value of new policies (numerical, float).
- **net_income**: Agent’s net income for the month, likely in monetary units (numerical, float).
- **number_of_policy_holders**: Current number of active policyholders linked to the agent (numerical, integer).
- **number_of_cash_payment_policies**: Number of policies paid using cash (numerical, integer).

0.4 Clarification of Units and Data Types

Most features in the dataset are numerical and represent counts or monetary values, with the exception of identifiers (e.g., Row ID, agent_code) and temporal information (e.g., year_month, agent_join_month). Count-based features are integers, while financial metrics like ANBP_value and net_income are floats. Categorical features such as month fields are encoded numerically, but represent discrete temporal categories.

0.5 Target Variable

The target variable is a binary label derived from the new_policy_count feature. If an agent sold at least one policy in a given month, they are labeled as 1 (Active); if no policies were sold, they are labeled as 0 (NILL). This target is critical to the business objective: predicting which agents are likely to become One Month NILL so that the company can intervene with support strategies to prevent disengagement or performance drop-off.

1 Key Metrics and Distribution Patterns

1.1 Key Metrics (Summary Statistics)

- `agent_age`:
 - Ranges mostly from early 20s to 60s.
 - Mean appears to be in the mid-30s to 40s.
- `unique_proposals` and its time splits (`last_7_days`, `last_15_days`, `last_21_days`):
 - Most agents submitted very few proposals.
 - A long right tail — few agents submitted a high number.
- `unique_quotations` and its splits:
 - Similar distribution to proposals — mostly low values, with a few high outliers.
- `unique_customers` and its splits:
 - Strongly right-skewed; majority of agents contact very few customers.
- `new_policy_count`:
 - The majority of agents have low counts.
 - Many zeros (NILL agents), justifying its use as a target for classification.
- `ANBP_value` and `net_income`:
 - Skewed distributions; most agents earn small amounts.
 - Some agents have very high values (outliers).
- `number_of_policy_holders`:
 - Highly skewed; majority of agents manage only a few policyholders.

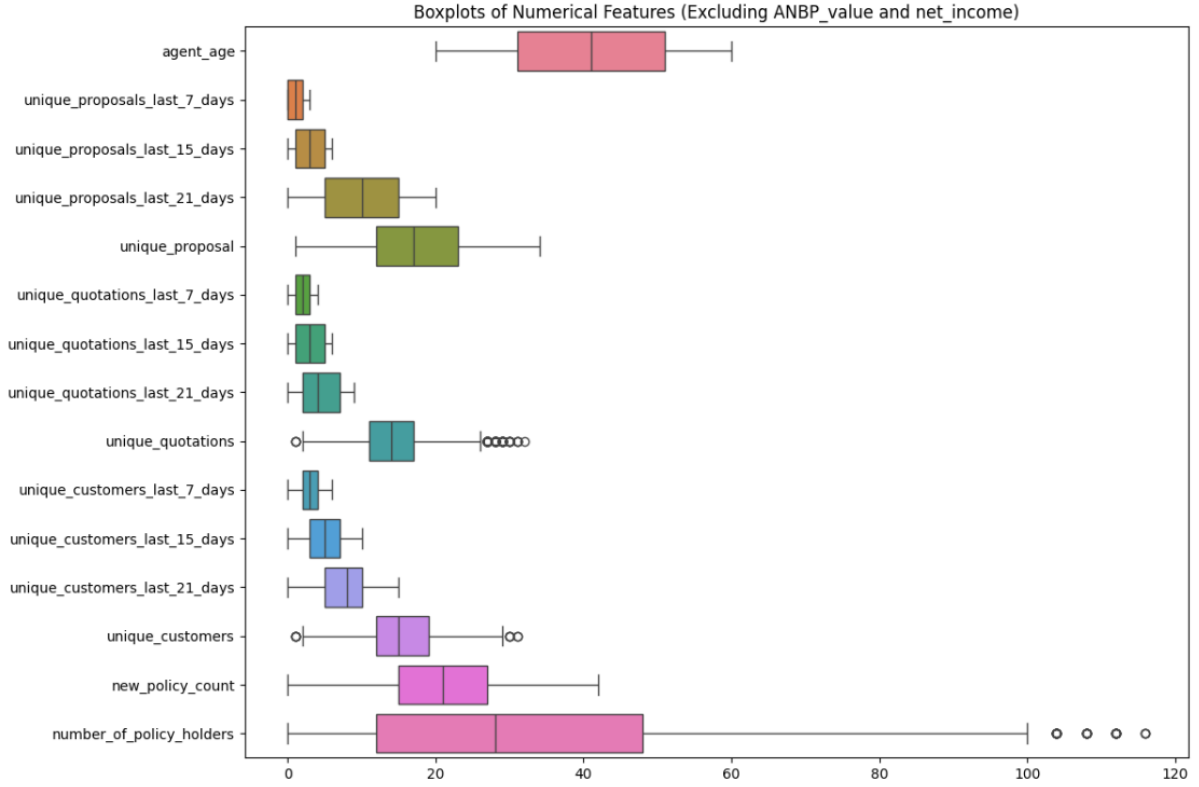


Figure 2: Box-Plots of Numerical Features

Table 1: Descriptive Statistics of Dataset Features

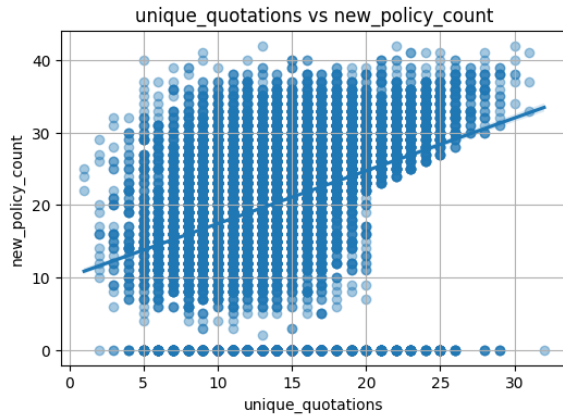
Feature	Count	Mean	Std	Min	25%	50%	75%	Max	Range	IQR
agent_age	15308	40.59	11.69	20.0	31.00	41.0	51.00	60.0	40.0	20.00
unique_prop_7d	15308	1.49	1.12	0.0	0.00	1.0	2.00	3.0	3.0	2.00
unique_prop_15d	15308	2.99	2.00	0.0	1.00	3.0	5.00	6.0	6.0	4.00
unique_prop_21d	15308	10.05	6.04	0.0	5.00	10.0	15.00	20.0	20.0	10.00
unique_proposal	15308	17.52	6.62	1.0	12.00	17.0	23.00	34.0	33.0	11.00
unique_quot_7d	15308	1.99	1.42	0.0	1.00	2.0	3.00	4.0	4.0	2.00
unique_quot_15d	15308	3.00	2.00	0.0	1.00	3.0	5.00	6.0	6.0	4.00
unique_quot_21d	15308	4.47	2.88	0.0	2.00	4.0	7.00	9.0	9.0	5.00
unique_quotations	15308	13.92	4.66	1.0	11.00	14.0	17.00	32.0	31.0	6.00
unique_cust_7d	15308	3.00	1.64	0.0	2.00	3.0	4.00	6.0	6.0	2.00
unique_cust_15d	15308	5.00	2.44	0.0	3.00	5.0	7.00	10.0	10.0	4.00
unique_cust_21d	15308	7.50	3.47	0.0	5.00	8.0	10.00	15.0	15.0	5.00
unique_customers	15308	15.49	4.54	1.0	12.00	15.0	19.00	31.0	30.0	7.00
new_policy_count	15308	20.27	9.54	0.0	15.00	21.0	27.00	42.0	42.0	12.00
ANBP_value*	15308	1025.3	805.5	0.0	347.3	903.1	1560.7	3933.8	3933.8	1213.4
net_income*	15308	228.0	173.7	1.2	92.8	190.0	321.7	1140.2	1139.0	228.9
policy_holders	15308	30.97	24.62	0.0	12.00	28.0	48.00	116.0	116.0	36.00
cash_payment_policies	15308	101.18	75.32	0.0	38.00	90.0	152.00	378.0	378.0	114.00

*Values for ANBP_value and net_income are shown in thousands

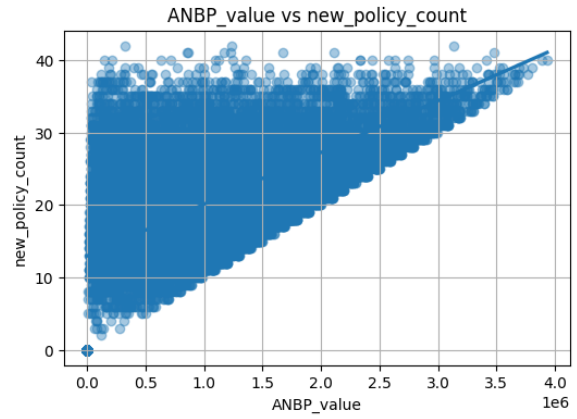
1.2 Distribution Patterns Observed

- All activity-based features (proposals, quotations, customer counts) show **right-skewed distributions**.
- Very few agents are highly active; most are moderately or minimally active.
- Target variable (NILL vs non-NILL) is **imbalanced** — more agents sell at least one policy than those who sell none.
- Some features (like `net_income`, `ANBP_value`, `unique_customers`) show **visible separation** when grouped by the target variable — indicating potential predictive power.

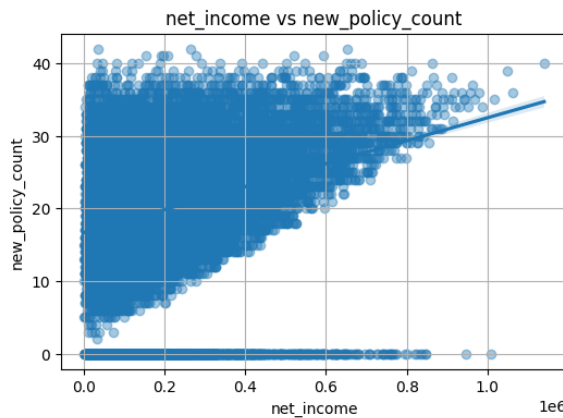
1.3 Some Patterns Observed



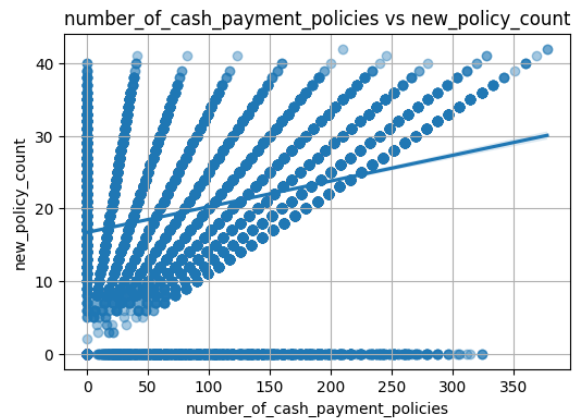
(a) Unique Quotations Vs New Policy Count



(b) ANPB Value Vs New Policy Count

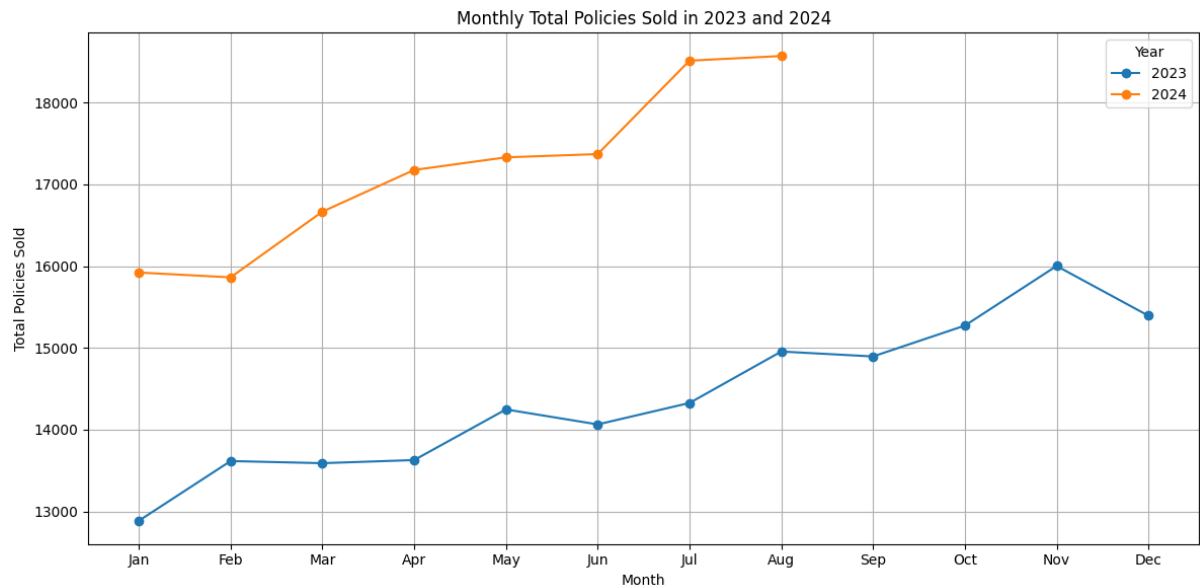


(a) Net Income Vs New Policy Count



(b) Number of Cash Payments Vs New Policy Count

2 Monthly Sales patterns variation



2.1 Monthly Policy Sales Analysis (2023 vs 2024)

The line chart compares the total number of policies sold each month in 2023 and 2024. A clear upward trend is observed in both years, with 2024 consistently outperforming 2023 in every month for which data is available.

2.1.1 2023 Sales Trends

- Sales started at around **12,900** in January and generally increased over the year.
- A **steady climb** can be seen from February through November, peaking at **16,000 policies** in November.
- A slight **drop is noticeable in December**, where sales fell back to around **15,400**—possibly due to seasonal slowdown or holiday effects.

2.1.2 2024 Sales Trends

- Sales began strong at **15,900** in January and rose steadily each month.
- There's a **significant spike between June and July**, jumping from ~17,400 to **18,500**.
- This peak continued into August, with sales reaching the year's high of **18,600**.
- The growth appears **more consistent and aggressive** than in 2023, with no noticeable declines in the available data range.

2.1.3 Key Observations

- **No unexpected drops** in 2024; the growth trajectory is stable and upward.

- **December 2023** is the only month showing a **minor decline**, potentially due to end-of-year factors.
- The **mid-year acceleration** in 2024 (June–August) suggests a possible campaign, market push, or seasonal demand surge.

3 Multivariate Analysis

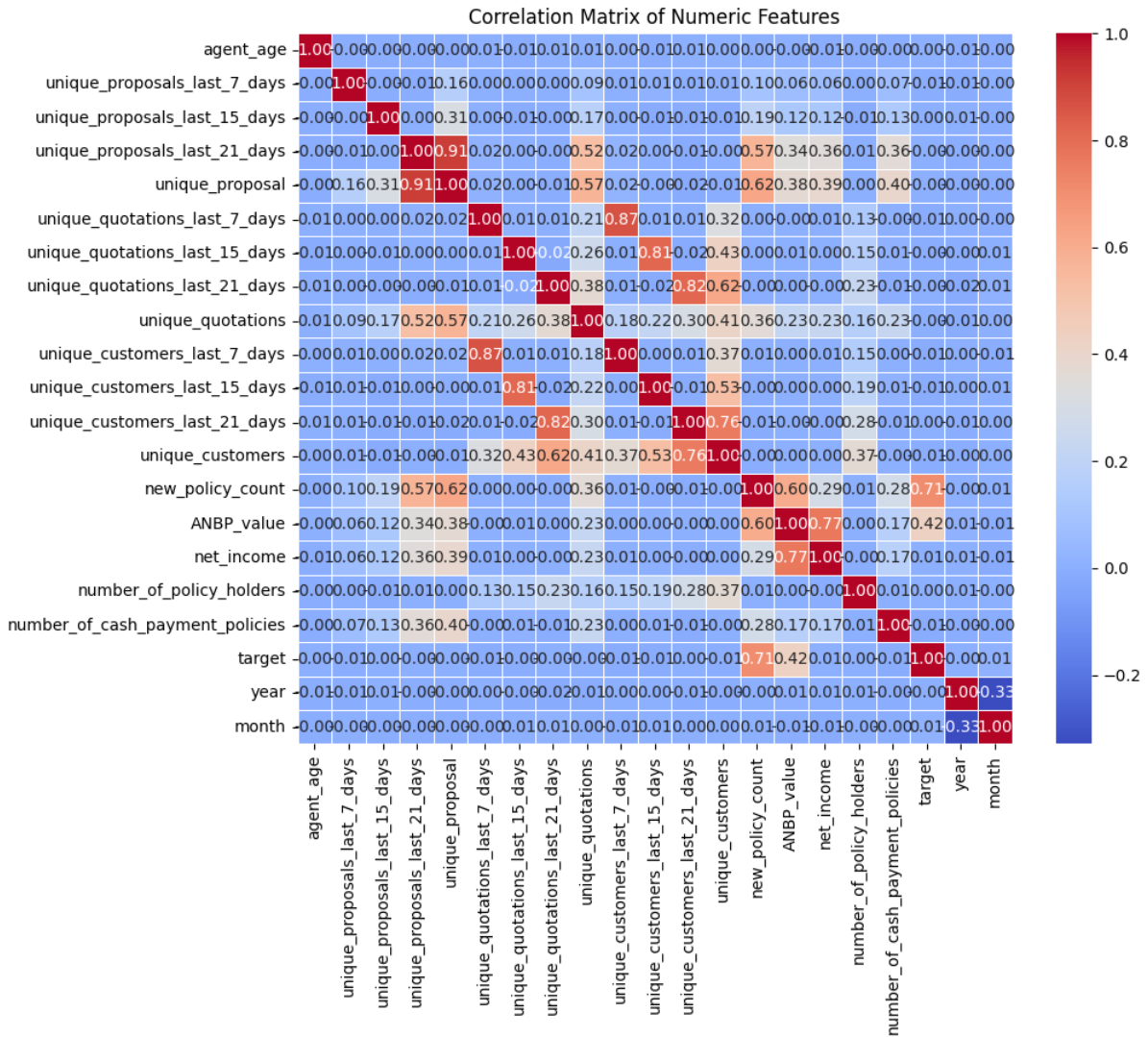


Figure 3: Correlation Diagram

3.1 Correlation Matrix Insights

Based on the correlation heatmap, here are the key insights derived from the numerical features:

3.1.1 Highly Correlated Feature Clusters

Several groups of features exhibit strong **positive correlations**, indicating they likely represent the same underlying information and could be redundant:

Proposals

- `unique_proposals_last_7/15/21_days` show high correlation with `unique_proposal` (**Correlation $\sim 0.57\text{--}0.91$**)
- These are expected to be subsets of the total proposal count, so the strong correlation is natural and highlights redundancy.

Quotations

- `unique_quotations_last_7/15/21_days` are strongly correlated with `unique_quotations` (**Correlation $\sim 0.82\text{--}0.88$**)
- Similar to the proposals, these are just segmented views of the same feature across different time windows.

Customers

- `unique_customers_last_7/15/21_days` are also strongly correlated with `unique_customers` (**Correlation $\sim 0.62\text{--}0.76$**)
- Again, these time-based features are subsets of the overall customer count.

Business Metrics

- `new_policy_count`, `ANBP_value`, and `net_income` show moderate correlations with:
 - `unique_customers`
 - `unique_proposals` (**Correlation $\sim 0.28\text{--}0.71$**)
- These metrics likely follow the same business trend—more customers or proposals tend to drive policy count and revenue.

3.1.2 Weak or No Correlation with Target

- The `target` variable shows **very weak correlation** with all other numerical features.
- This indicates that **linear relationships** do not explain the target well, and more complex (nonlinear) models may be needed to capture useful patterns.

Agent Age

- `agent_age` appears **uncorrelated** with all other features, including `target`.
- This suggests it may not contribute much predictive value on its own, though it might still be useful in combination with other features.

3.2 Temporal Features

- month has a **mild correlation** (~ 0.33) with **target**, suggesting some **seasonal trends** could be present.
- year shows **negligible correlation**, likely due to limited variation (e.g., data collected within a single year).

3.3 Summary

- Time-windowed features are heavily correlated with their cumulative counterparts—consider aggregating or dropping some.
- Nonlinear models are likely better suited for predicting the target.
- Look into feature interactions or temporal patterns (e.g., seasonal effects) for additional insight.

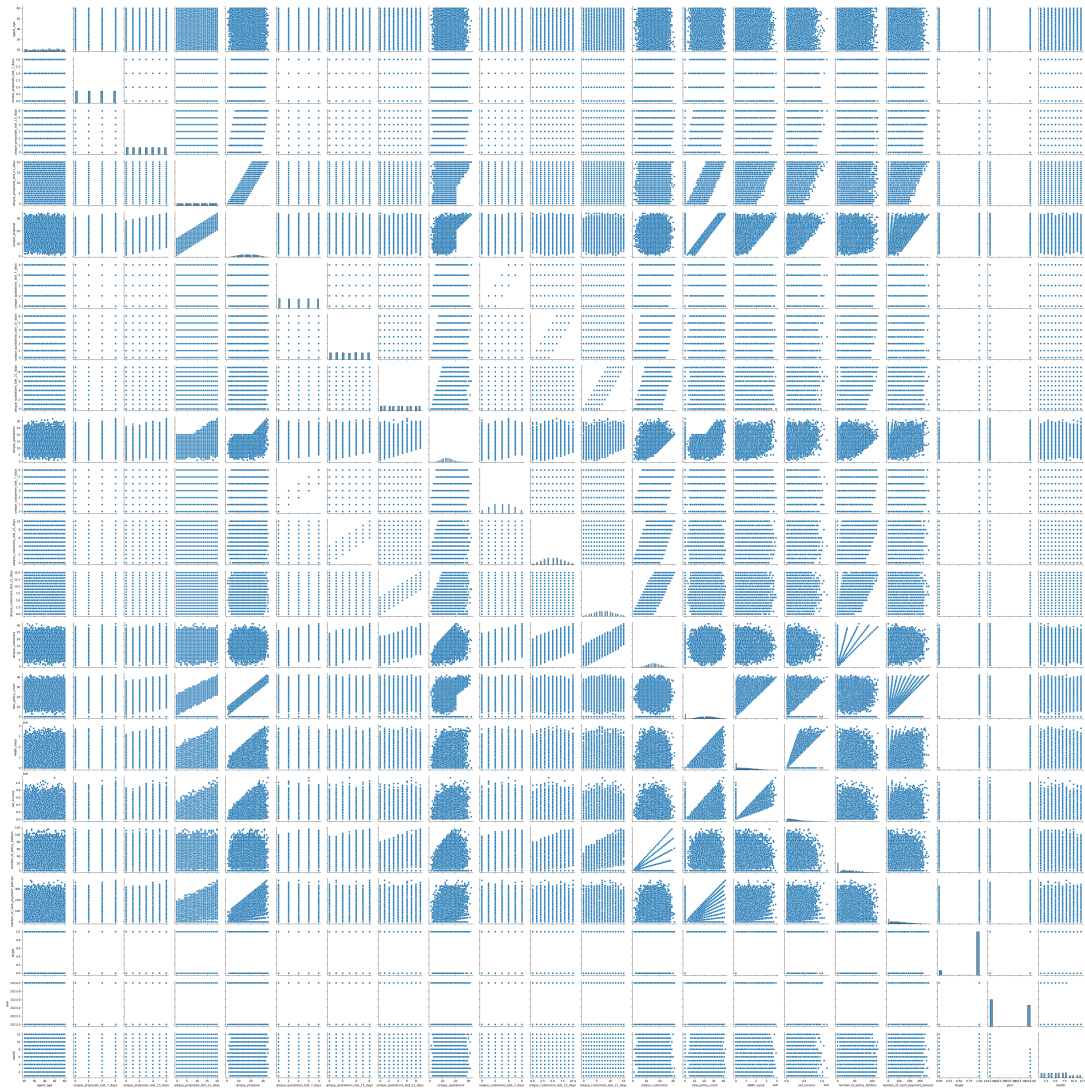
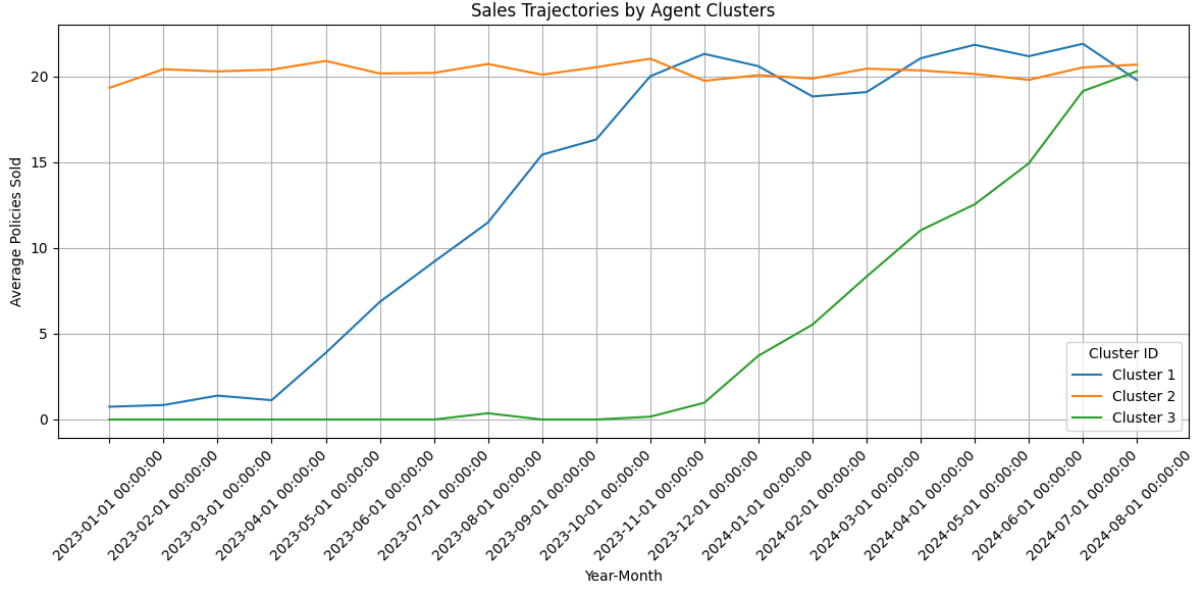


Figure 4: Pairplot for the Dataset

4 Agents' Performance Evolution



4.1 Performance Analysis of Clusters

- **Cluster 1 (High Performers):** Agents in Cluster 1 consistently performed well throughout the study period. They maintained a high level of performance with a stable average of policies sold, showing steady results over time.
- **Cluster 2 and Cluster 3 (Beginners):** Both Cluster 2 and Cluster 3 consist of beginner agents with a significantly lower average of policies sold compared to Cluster 1. However, with time, both clusters managed to improve and eventually reached the performance level of Cluster 1.
- **Cluster 3 - Slower Progress:** While Cluster 2 agents were able to catch up with the performance of Cluster 1 relatively quickly, Cluster 3 agents took considerably longer to reach the same level of success.

5 Innovative EDA

5.1 Behavioral Consistency Score

What is it?

The Behavioral Consistency Score is a custom metric designed to measure how stable and predictable an agent's performance is over time.

Why it matters?

Agents who perform consistently across months—in terms of proposals, quotations, customers, and income—may be more reliable, easier to manage, and better suited for long-term planning compared to agents with high volatility.

How is it calculated?

1. Select Key Metrics:

We use the following performance indicators:

- new_policy_count
- unique_proposal
- unique_quotations
- unique_customers
- ANBP_value
- net_income

2. Group by Agent and Month:

Aggregate the monthly performance of each agent.

3. Rolling Coefficient of Variation (CV):

For each agent, compute the CV of each metric over a rolling 3-month window:

$$CV = \frac{\text{Standard Deviation}}{\text{Mean}}$$

A lower CV indicates more stable performance.

4. Aggregate CVs to Compute Consistency:

Average the CVs across all metrics and take the inverse to obtain the consistency score:

$$\text{Consistency Score} = \frac{1}{\text{Average CV} + \epsilon}$$

where ϵ is a small value to avoid division by zero.

Interpretation:

Higher score $\Rightarrow$ More behaviorally consistent (stable)

Lower score $\Rightarrow$ More volatile or irregular performance

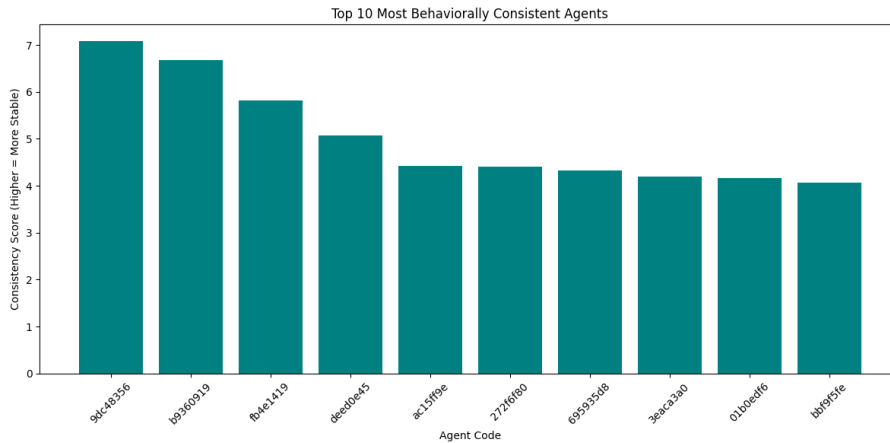


Figure 5: Top 10 Most Behaviorally Consistent Agents

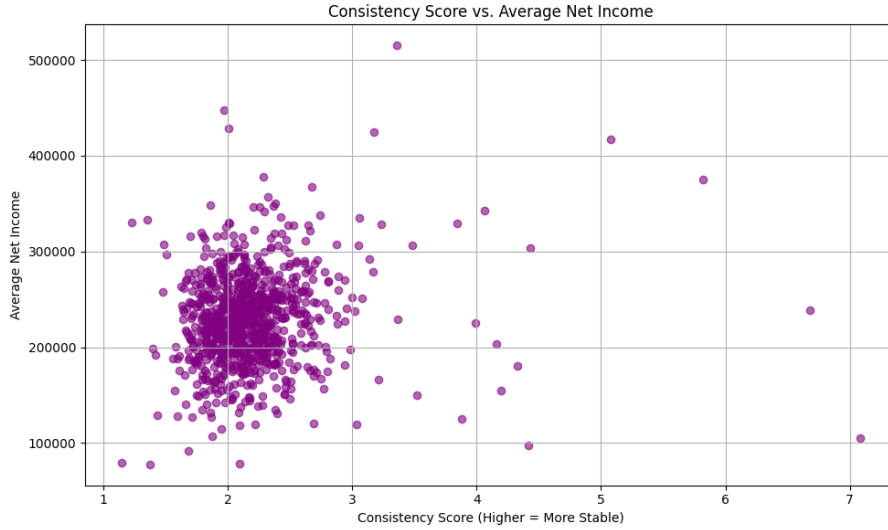


Figure 6: Scatter plot of Consistency Score vs. Average Net Income

5.2 Insights: Consistency Score vs. Average Net Income

- **Tight Cluster Around Consistency Score ~2:** Most agents have scores between 1.5 and 2.5, with average net incomes clustering around 200K–250K.
- **Higher Consistency $\nRightarrow$ Guaranteed Higher Income:** A few agents with high consistency (>4) have mixed income levels.
- **Presence of Outliers:** Some low-consistency agents earn very high income (likely volatile high performers), while some high-consistency agents earn less (possibly underleveraged).
- **Key Insight:** Moderate consistency is most common and aligns best with income stability. There is no clear linear correlation between consistency and income, suggesting multiple successful agent profiles.

5.3 Insights from Ramp-up Time Distribution

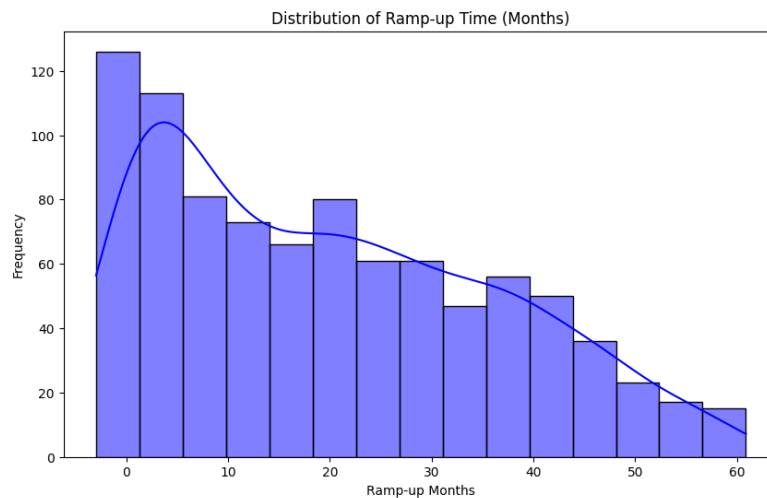


Figure 7: Distribution of Ramp-up Time Across Agents

- Most agents sell their first policy within the first 0–5 months.
- Number of agents steadily declines with increased ramp-up time.
- A long tail exists: some agents take more than 12 months, even up to 60 months.
- **Implication:** Early onboarding/training effectiveness plays a major role in initial success.

5.4 Insights from Peak Performance by Ramp-up Category

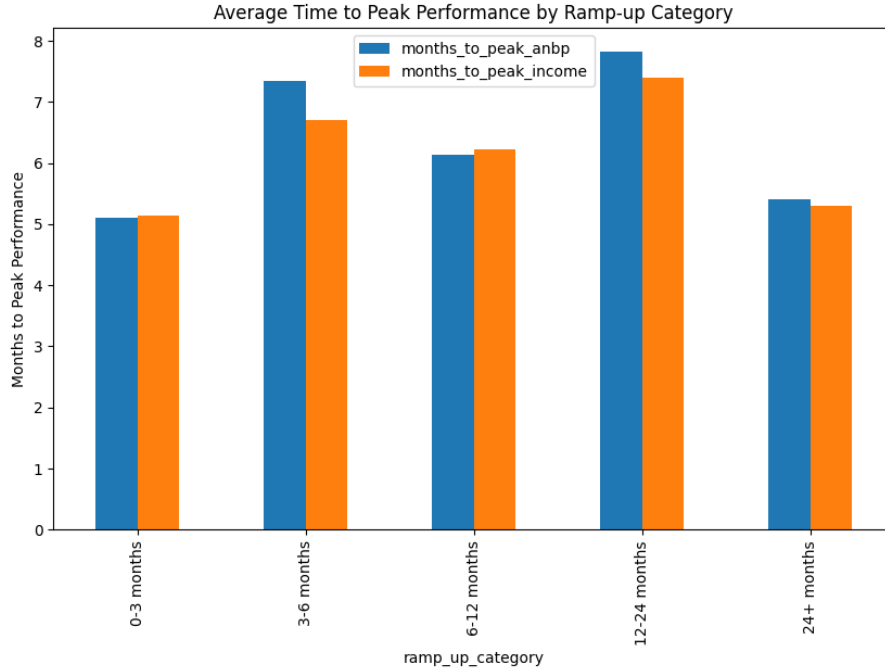


Figure 8: Average Time to Peak Performance by Ramp-up Category

- **Fast ramp-up, fast peak:** Agents who ramp up within 0–3 months reach peak performance fastest (~5.1 months).
- **Delayed ramp-up, slower peak:** Agents ramping up in 12–24 months take longest to peak (~7.8 months).
- **Moderate groups vary:** 3–6 month ramp-up agents take longer to peak than others.
- **Interesting outlier:** >24 month ramp-up group peaks faster than some mid-range groups.
- **Consistent patterns:** Peak timing for ANBP and income metrics follow similar trends across ramp-up categories.

Chapter 2 - NILL Agents Prediction

1 Prediction on the Agents

1.1 Introduction

The model designed to predict whether an insurance agent will sell policies in the next month based on historical data. It uses a binary classification approach with the XGBoost algorithm, performing data preprocessing, hyperparameter tuning, model training, and feature importance visualization.

1.2 Data Preprocessing

- **Input:** Loads dataset from `train_storming_round.csv`.
- **Date Handling:**
 - Converts `year_month` to datetime format.
 - Extracts separate `year` and `month` features.
- **Target Creation:**
 - Creates binary target variable `target`, where:
 - * 1 indicates the agent sells at least one policy in the next month.
 - * 0 indicates no policies sold (null).
 - Uses `shift(-1)` to get next month's policy count per agent.
 - Drops rows where target is unavailable (last month per agent).
- **Feature Selection:**
 - Excludes columns that could cause data leakage:
 - * `month`, `policies_sold`, `next_month_policies`, `target`.
 - Removes non-numeric or non-datetime columns (e.g., `row_id`).

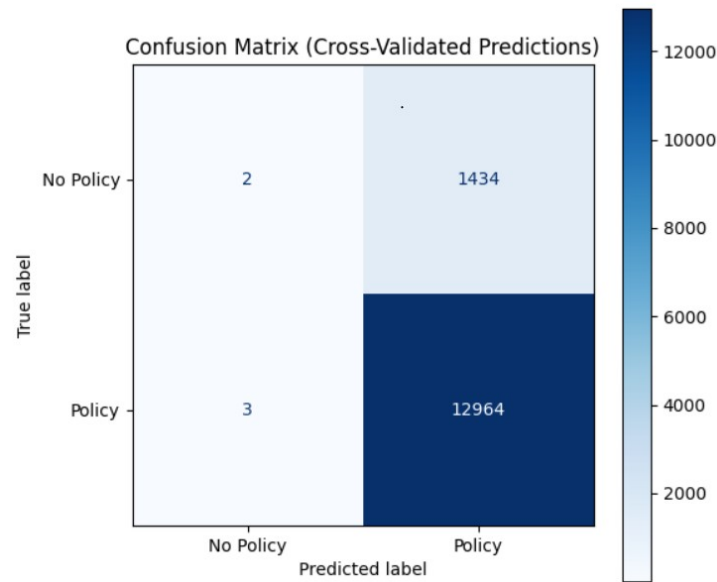
1.3 Hyperparameter Tuning

- Uses Optuna for optimization to maximize the F1 score.
- **Tuned Parameters:**
 - `learning_rate`: 0.01 to 0.2
 - `max_depth`: 3 to 10

- `n_estimators`: 100 to 600
- `subsample`: 0.6 to 1.0
- `colsample_bytree`: 0.6 to 1.0
- `gamma`: 0 to 1.0
- `reg_lambda`: 0.1 to 10.0
- Performs 5-fold stratified cross-validation for evaluation.
- Outputs best F1 score and corresponding parameters.

1.4 Model Training

- Trains an `XGBoostClassifier` using best parameters from Optuna.
- Uses preprocessed dataset (`X`, `y`) for training.



1.5 Results

- It was able to secure a F1 score of 0.94460 in the Kaggle Space.

1.5.1 F1 Score

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

2 Powerful factors

2.1 Feature Importance Visualization

- Used `plot_importance` from XGBoost with `importance_type='gain'`.
- Displays a bar plot showing each feature's contribution to the model's prediction.

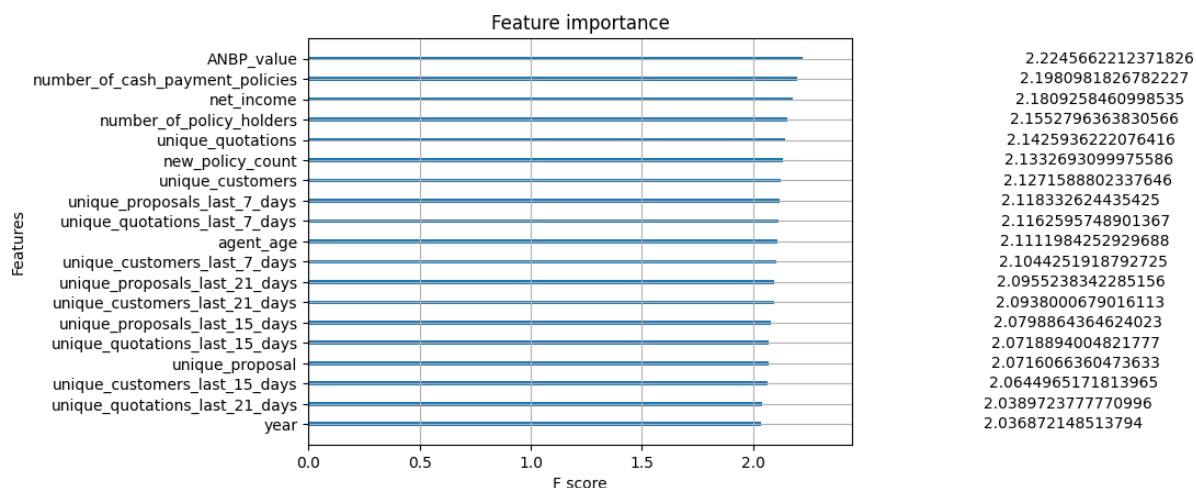


Figure 9: Feature Importance Diagram

2.2 Top Features (Highest Weight to Lowest)

1. ANBP_value
2. number_of_cash_payment_policies
3. net_income
4. number_of_policy_holders
5. unique_quotations

3 Action Plans for At-Risk Agents

3.1 Introduction

1. Personalized Coaching Recommendation System for At-Risk Agents This system is designed to proactively identify insurance agents who are at risk of underperforming and generate tailored coaching plans to help them improve. It leverages the machine learning model trained to predict NILL agents The workflow is as follows:
2. Data Preparation: Test data is preprocessed to extract relevant features while removing columns that should not influence predictions, such as historical sales or identifiers.

3. Risk Prediction: Using the pre-trained model (model.pkl), the system computes the probability of each agent meeting their next-month policy sales goal. Agents with a probability below a defined threshold (e.g., 0.7) are flagged as "at risk."
4. Explainability with SHAP: For each at-risk agent, SHAP (SHapley Additive ex-Planations) values are computed to understand which features contributed most to their risk score. The top three contributing features are identified as key weaknesses or improvement areas.
5. Personalized Coaching Plan Generation: Each of these top features is mapped to a predefined coaching strategy. These strategies include actionable advice such as improving proposal timing, expanding customer outreach, or transitioning to digital payment methods. If a feature has no associated plan, a default message is included.
6. Output: The system generates a final report listing each at-risk agent, their risk score, the top three reasons for their risk status, and a personalized coaching plan. This output is saved as agent_coaching_plans.csv, providing managers with clear, data-driven guidance for targeted mentoring.

3.2 Feature-wise Coaching Strategies

- **net_income:** Provide personalized coaching to help the agent identify and pursue higher-value opportunities. Review past sales to identify missed income potential.
- **ANBP_value:** Coach the agent on improving business quality and premium value by focusing on client needs and cross-selling opportunities.
- **unique_proposals_last_21_days:** Work with the agent on optimizing proposal timing and targeting, ensuring consistency in lead follow-up and conversion strategy.
- **agent_age:** Offer age-specific mentoring—young agents may benefit from product knowledge training, while older ones might need digital tool support.
- **number_of_cash_payment_policies:** Educate the agent on the risks and best practices of cash handling, and promote digital payment solutions for better client tracking.
- **number_of_policy_holders:** Support the agent in building long-term client relationships and retention strategies to grow a loyal customer base.
- **unique_proposals_last_15_days:** Guide the agent to review proposal success rates and fine-tune their approach to ensure productive outreach.
- **unique_quotations:** Mentor on quote effectiveness—how to target the right clients and ensure quotations are tailored and likely to convert.
- **unique_customers_last_15_days:** Encourage the agent to increase and diversify outreach, and provide tools for efficient prospecting in new segments.
- **unique_proposal:** Review historical sales cycles with the agent to reignite consistent proposal generation and uncover potential bottlenecks.
- **unique_customers:** Mentor the agent on portfolio diversification and expanding customer reach, using referral strategies and community engagement.

- **unique_customers_last_21_days:** Re-engage agent in consistent customer interactions and offer time-management coaching to maintain relationship momentum.
- **new_policy_count:** Provide coaching on sustainable business practices and verifying client intent to avoid short-term gain but long-term loss.
- **unique_quotations_last_7_days:** Help the agent reflect on recent quotation patterns and adjust for better targeting and time efficiency.
- **unique_proposals_last_7_days:** Offer real-time coaching to improve proposal quality and address any rushed or repetitive approaches.
- **unique_quotations_last_21_days:** Collaborate with the agent on streamlining the quoting process, ensuring follow-through and effective time use.
- **unique_quotations_last_15_days:** Provide coaching sessions focused on improving recent quotation outcomes and understanding client objections.
- **unique_customers_last_7_days:** Help the agent build consistent weekly routines to ensure they stay connected with a healthy mix of prospects and clients.

3.3 Agent 27f35403 (Risk Score: 0.65)

3.3.1 Contributing Factors

- **Unique Proposals Last 21 Days** (SHAP: ↓0.5047): Low proposal activity in the last 21 days significantly reduces the likelihood of sales.
- **Unique Proposals Last 15 Days** (SHAP: ↓0.2713): Recent lack of proposal engagement further contributes to the risk.
- **Agent Age** (SHAP: ↓0.1942): Age-related challenges may be hindering performance.

3.3.2 Personalized Action Plan

- Work with the agent on optimizing proposal timing and targeting, ensuring consistency in lead follow-up and conversion strategy.
- Guide the agent to review proposal success rates and fine-tune their approach to ensure productive outreach.
- Offer age-specific mentoring—young agents may benefit from product knowledge training, while older ones might need digital tool support.

3.4 Agent f2fcfe25 (Risk Score: 0.68)

3.4.1 Contributing Factors

- **Agent Age** (SHAP: ↓0.4285): Age-related barriers are the primary contributor to the risk.
- **Net Income** (SHAP: ↓0.3615): Low income indicates missed high-value opportunities.
- **Unique Quotations** (SHAP: ↓0.3215): Ineffective or low quotation activity reduces sales potential.

3.4.2 Personalized Action Plan

- Offer age-specific mentoring—young agents may benefit from product knowledge training, while older ones might need digital tool support.
- Provide personalized coaching to help the agent identify and pursue higher-value opportunities. Review past sales to identify missed income potential.
- Mentor on quote effectiveness—how to target the right clients and ensure quotations are tailored and likely to convert.

3.5 Agent 6452bd08 (Risk Score: 0.68)

3.5.1 Contributing Factors

- **Net Income** (SHAP: ↓0.3143): Low income suggests missed high-value sales opportunities.
- **Unique Proposals Last 21 Days** (SHAP: ↓0.2142): Reduced proposal activity indicates weak lead generation.
- **Number of Policy Holders** (SHAP: ↓0.2118): A small policyholder base limits retention and growth.

3.5.2 Personalized Action Plan

- Provide personalized coaching to help the agent identify and pursue higher-value opportunities. Review past sales to identify missed income potential.
- Work with the agent on optimizing proposal timing and targeting, ensuring consistency in lead follow-up and conversion strategy.
- Support the agent in building long-term client relationships and retention strategies to grow a loyal customer base.

3.6 Agent 60b8427e (Risk Score: 0.63)

3.6.1 Contributing Factors

- **ANBP Value** (SHAP: ↓0.4097): Low Annual Net Business Premium value indicates poor business quality.
- **Net Income** (SHAP: ↓0.2575): Low income reflects missed sales opportunities.
- **Number of Policy Holders** (SHAP: ↓0.2227): A small policyholder base hinders growth.

3.6.2 Personalized Action Plan

- Coach the agent on improving business quality and premium value by focusing on client needs and cross-selling opportunities.
- Provide personalized coaching to help the agent identify and pursue higher-value opportunities. Review past sales to identify missed income potential.

- Support the agent in building long-term client relationships and retention strategies to grow a loyal customer base.

3.7 Agent 6c9d3c40 (Risk Score: 0.64)

3.7.1 Contributing Factors

- **ANBP Value** (SHAP: ↓0.3403): Low premium value suggests poor business quality.
- **Net Income** (SHAP: ↓0.1916): Low income indicates inefficiencies in sales.
- **Unique Proposals Last 15 Days** (SHAP: ↓0.1641): Recent lack of proposals reduces sales potential.

3.7.2 Personalized Action Plan

- Coach the agent on improving business quality and premium value by focusing on client needs and cross-selling opportunities.
- Provide personalized coaching to help the agent identify and pursue higher-value opportunities. Review past sales to identify missed income potential.
- Guide the agent to review proposal success rates and fine-tune their approach to ensure productive outreach.

3.8 Agent d360368b (Risk Score: 0.59)

3.8.1 Contributing Factors

- **Unique Customers Last 15 Days** (SHAP: ↓0.3286): Limited customer engagement reduces sales opportunities.
- **New Policy Count** (SHAP: ↓0.3272): Low new policy counts indicate ineffective sales practices.
- **Unique Quotations** (SHAP: ↓0.2486): Poor quotation activity limits conversions.

3.8.2 Personalized Action Plan

- Encourage the agent to increase and diversify outreach, and provide tools for efficient prospecting in new segments.
- Provide coaching on sustainable business practices and verifying client intent to avoid short-term gain but long-term loss.
- Mentor on quote effectiveness—how to target the right clients and ensure quotations are tailored and likely to convert.

3.9 Agent f7136abb (Risk Score: 0.52)

3.9.1 Contributing Factors

- **Number of Cash Payment Policies** (SHAP: ↓0.5020): High reliance on cash payments may indicate inefficiencies.

- **Net Income** (SHAP: ↓0.3602): Low income suggests missed high-value opportunities.
- **Unique Proposals Last 21 Days** (SHAP: ↓0.2976): Low proposal activity reduces sales potential.

3.9.2 Personalized Action Plan

- Educate the agent on the risks and best practices of cash handling, and promote digital payment solutions for better client tracking.
- Provide personalized coaching to help the agent identify and pursue higher-value opportunities. Review past sales to identify missed income potential.
- Work with the agent on optimizing proposal timing and targeting, ensuring consistency in lead follow-up and conversion strategy.

3.10 Agent 5724f683 (Risk Score: 0.56)

3.10.1 Contributing Factors

- **Unique Proposals Last 21 Days** (SHAP: ↓0.2372): Low proposal activity is a key risk factor.
- **Number of Cash Payment Policies** (SHAP: ↓0.2050): Cash payment reliance may cause inefficiencies.
- **Net Income** (SHAP: ↓0.2034): Low income indicates missed opportunities.

3.10.2 Personalized Action Plan

- Work with the agent on optimizing proposal timing and targeting, ensuring consistency in lead follow-up and conversion strategy.
- Educate the agent on the risks and best practices of cash handling, and promote digital payment solutions for better client tracking.
- Provide personalized coaching to help the agent identify and pursue higher-value opportunities. Review past sales to identify missed income potential.

4 Conclusion

These action plans are designed to address the specific performance gaps of each at-risk agent, leveraging insights from predictive modeling and SHAP analysis. By implementing these SMART interventions, management can support agents in improving their sales performance and preventing One Month NILL. Regular monitoring of progress (e.g., proposal conversion rates, income growth) is recommended to evaluate the effectiveness of these plans.

Chapter 3 - Monitoring and Improving Existing Agent Performances

1 K-means Clustering Process

1.1 Introduction

To evaluate and enhance the performance of current agents, we conducted a comprehensive analysis of the historical dataset. Using the `groupby` and `agg()` functions in Python, we calculated key agent performance metrics. These included the total, average, maximum, and variability (standard deviation) of policies sold, providing insight into sales consistency and peak performance.

In addition to sales-related data, we incorporated engagement metrics such as the number of unique proposals, customers, and quotations handled by each agent. This multi-dimensional approach enabled us to capture both quantitative output and customer engagement trends, laying the foundation for categorizing agents and designing targeted improvement strategies.

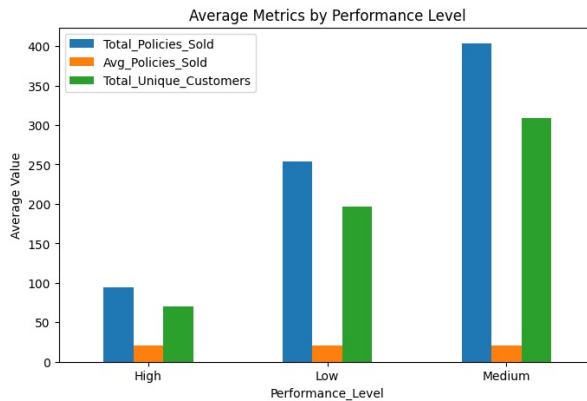


Figure 10: Avg Metrics by Performance Level

1.2 Performance Metrics Used

We aggregated multiple performance indicators for each agent, including:

- Total Policies Sold
- Average Policies Sold per Record
- Standard Deviation and Maximum of Policies Sold
- Total Unique Proposals
- Total Unique Customers
- Total Unique Quotations

These metrics provide a balanced view of consistency, peak performance, and customer/proposal engagement.

1.3 Clustering and Classification

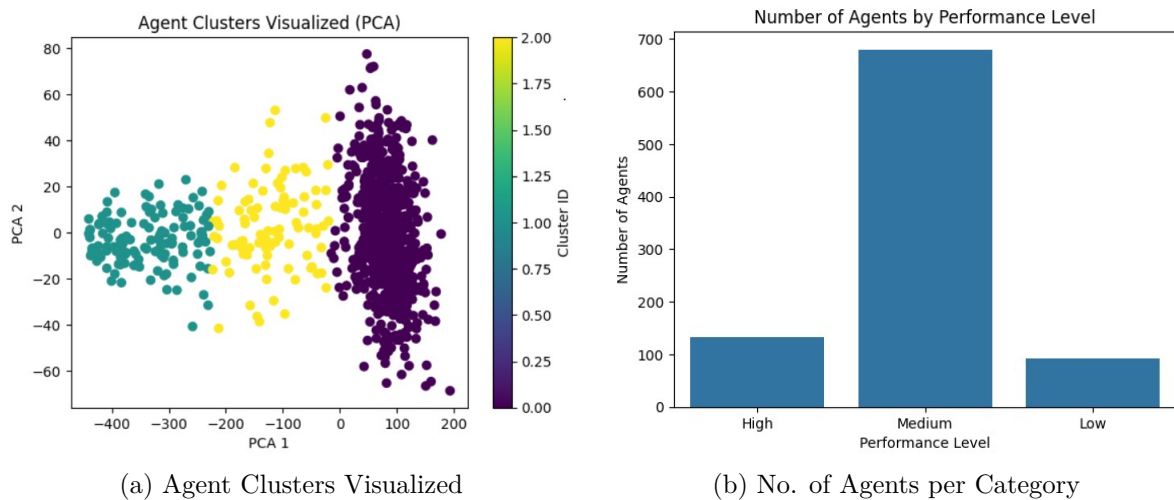
Using the K-Means clustering algorithm, agents were segmented into three groups:

- **High Performers**
- **Medium Performers**
- **Low Performers**

The clustering was based on six key features. The clusters were labeled by sorting the cluster centers based on average policies sold.

Cluster Label	Description
High	Consistently strong performers across most metrics
Medium	Average performance, with room for growth
Low	Underperformers needing significant support

Table 2: Agent Performance Clusters



1.4 Recommendations for Intervention

Performance Level	Suggested Interventions
High	Offer incentives (bonuses, recognition), assign them as mentors to others, provide autonomy in choosing leads.
Medium	Provide additional training, regular performance reviews, and set incremental sales goals.
Low	Conduct one-on-one coaching, reduce administrative burden, and provide simplified sales materials.

Table 3: Recommended Interventions by Performance Level

2 Tracking Progress

Each agent's cluster label is saved and can be tracked over time. By repeating this clustering process periodically, we can:

- Detect changes in agent performance levels
- Assess whether interventions are effective
- Identify rising or declining performers early

A dashboard has been created and all the relative files are attached externally with a README file to show predictions, classifications, recommendations, and to make Progress tracking easy.